

Review article

A review of polymethyl methacrylate (PMMA) as a versatile lithographic resist – With emphasis on UV exposure

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ABSTRACT

This review describes the utility of polymethyl methacrylate (PMMA) as a resist for several types of lithographic processes. Ordinarily, PMMA is only considered an electron beam resist but in fact it can be used as a resist material with exposure systems based on both particle beams and electromagnetic radiations. In each case, both positive tone and negative tone resist action can be realized with appropriate process design. We first describe and review the mechanisms that bring about lithographic pattern formation in this acrylic polymer. This is followed by descriptions of processes for using PMMA in different ways as a lithographic resist. Use of PMMA at both long and short UV wavelengths is treated in more detail as this aspect is less well-known for lithography with PMMA. We also describe several new findings from own work here, such as cross-linking PMMA with Michler's ketone (MK), use of resist reflow for three-dimensional structure fabrication and lithographic tone switching with change in UV exposure dose. The aim of this exposition is to emphasize the great versatility of PMMA as a resist for micro- and nano-electronic device fabrication.

1. Introduction

Polymethyl methacrylate (PMMA) is a well-known resist for electron beam lithography [1,2]. In this role, it offers the highest resolution of any polymeric resist [3–6]. Higher resolutions have only been obtained in inorganic salt films used as resists but that approach is only of academic interest [7]. Sub-10 nm features have been produced in thin PMMA films through exposure with vector electron beam lithography systems [8–11]. Its high resolution originates from its relatively simple molecular structure and propensity for clean and facile chain scission when bombarded with energetic particles or photons. PMMA has been popular as an electron beam resist for many years due to its easy availability in several molecular weights, low cost, non-toxic environmentally-friendly character, ease of processing and wide literature coverage. While PMMA is widely used for electron beam lithography, its use with electromagnetic radiation is less well-known. X-ray lithography has been carried out with PMMA for a number of years [12,13] and even deep UV lithography has seen its use [14] but the utility of this polymer at longer UV wavelengths is much less known. As we show later in this paper, PMMA can be used as a resist at any wavelength,

below about 400 nm. In fact, PMMA even shows a positive tone to negative tone transition with increasing UV dose – similar to what is observed with electron beam lithography. These discussions highlight the versatility of PMMA as a patternable masking material. Not only can planar lithographic patterns be produced in PMMA, it is also possible to ‘reflow’ such patterns to create three-dimensional structures, albeit of a limited variety. Furthermore, we also show how the well-known weakness of PMMA against damage from dry etch plasmas can be overcome; at least to a limited extent. Our coverage includes a short description of PMMA's use as an electron beam resist in order to give context to our later discussion of combined electron beam / optical lithography. However, in this paper we mainly deal with PMMA's use with UV radiation. Additionally, we show that PMMA can also be used in lithographic processes where electron beam and UV radiation is applied to the same PMMA film, one after the other, creating both small and large features together. Further information can be obtained from other publications that describe different lithography processes in more detail. For an even broader coverage of both lithography and associated microfabrication techniques the review by del Campo and Arzt can be consulted [15].

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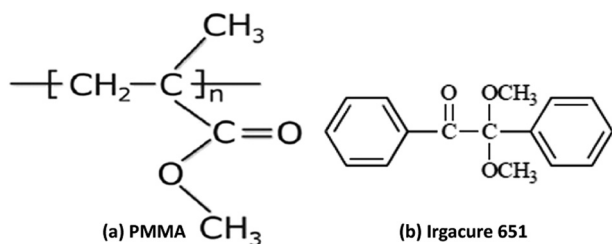


Fig. 1. (a) Chemical structure of the repeating unit of PMMA (b) Chemical structure of an Irgacure 651 molecule.

2. Lithographic behaviour of PMMA

2.1. Properties of PMMA

Part (a) of Fig. 1 shows the structure of the repeat unit of PMMA. The aliphatic $\text{CH}_2\text{-C}$ - backbone is connected to CH_3 - (methyl) and $\text{CH}_3\text{-O-CO-}$ (ester) side groups. The aliphatic (open chain) character of PMMA plays a very significant role in how this polymer behaves toward lithographic exposure and dry etching. As an acrylic polymer, PMMA belongs to the family of polymers that are formed by the polymerization of an alkyl acrylate ester – in this case, the ester of methyl alcohol with methacrylic acid i.e. methyl methacrylate. PMMA is generally synthesized from its liquid monomer – methyl methacrylate - through the use of a radical polymerization initiator, such as benzoyl peroxide. The reaction is exothermic and the temperature needs to be controlled for the polymerization to proceed properly. The duration of polymerization controls the molecular weight of the product formed and thus the length of PMMA chains obtained. Timed termination is commonly used to produce PMMA of different mean molecular weights. Once synthesized, and if required, PMMA can be purified by column-based gel chromatography systems to obtain a mono-dispersed batch with tightly-controlled molecular weight distribution. Commonly available variants have molecular weights ranging from less than 5000 g/mol to around 1000,000 g/mol. These products can be purchased, as powders or granules, from a number of chemical distributors, such as Sigma-Aldrich and Alfa Aesar. It is also possible to synthesize PMMA having molecular weight outside the range quoted above and this is sometimes done for specialized applications, such as in bio-medical sciences.

PMMA can be dissolved in a wide variety of organic solvents, both aliphatic and aromatic, such as toluene, xylene, dichloromethane, chlorobenzene, ethyl lactate, ethyl acetate etc. However, ethyl lactate is a particularly good solvent for PMMA because of its environmentally benign nature. It is non-toxic to humans and is also bio-degradable [16]. When made into a solution, typical PMMA concentrations range from 3% to 20%, (w/w). The exact value depending on the PMMA film thickness desired, with higher concentrations leading to thicker films. PMMA films can be spin-coated onto flat substrates from such solutions. Film thickness produced by spin-coating depends on both the concentration (solid content) of the PMMA solution used and the spin speed used. Small feature size, high resolution patterning makes use of thin PMMA films (typically 100 to 300 nm thick, depending on desired feature sizes and etch depths) whereas large feature size and/or high aspect ratio patterning makes use of thicker films (up to several microns in thickness, prepared through repeated spin coating). The vapour pressure of ethyl lactate is low enough to produce good quality uniform and striation-free films in this thickness range. A particularly good property of PMMA is that its solutions (even at high concentrations) do not have a tendency to gel through chain entanglement, as some other polymers, such as polycarbonate, do. Thus, PMMA solutions have, effectively, an un-limited shelf life.

Like other acrylic polymers, PMMA is also transparent to light. In fact, it is one of the most transparent polymers known, with transparency extending down to quite short UV wavelengths. Fig. 4 shows the

featureless transmission of PMMA in the long wavelength UV region. The tendency seen here continues down to around 300 nm. This property benefits PMMA's applications in lithography in several ways. Because of transparency at long wavelengths, underlying structures are clearly visible through a thin PMMA film, with both optical and scanning electron microscopes (SEMs), which makes it easy to carry out pattern alignment. Transparency to short wavelengths also makes it possible to use it as a deep UV (DUV) and X-ray resist. Furthermore, modern 248 nm and especially 193 nm resists are based on substituted forms of acrylic polymers on account of their outstanding transparency to wavelengths around 200 nm [17–20]. Most other polymers absorb too much at such short wavelengths so that sufficient radiation does not penetrate to lower portions of resist films in order to expose the film in a uniform manner. Thus, the behaviour of PMMA also informs us about the design of commercial DUV resists.

2.2. Mechanism of resist action in PMMA

The photochemistry of PMMA and related acrylic polymers is rich and interesting. Certain properties of the chemical structure of PMMA help make it amenable to act as a resist. Work by many research groups has shown that PMMA can be used as both a positive and a negative tone resist under appropriate circumstances. Of the two, positive tone lithographic action is both easier to produce and the more widely used. This comes about because of the ease with which PMMA molecular chains can fragment when exposed to sufficiently energetic electrons, ions or photons [21]. Mostly this happens through the shearing of the side groups from the backbone but in some cases the main backbone itself breaks. Through a host of techniques it has been shown that the ester side group of PMMA chain is most susceptible to shearing (see Fig. 1 (a) for the structure of the repeating unit of PMMA). These processes produce shorter chain fragments in exposed regions of PMMA film which can be preferentially dissolved in weak solvents, such as diluted methyl isobutyl ketone (MIBK) [22–24]. Especially good results can be achieved if the development is carried out slowly at low temperature [25–27]. Temperatures around -20°C slow down the dissolution rate of PMMA, avoiding over-development which almost always occurs with room temperature development. The resulting difference in film topography produces the lithographic image in the PMMA resist film. Together with chain scission, some cross-linking also takes place because some of the fragments produced in chain fragmentation are free radicals which cause secondary inter-chain and intra-chain bridging reactions. Thus, both chain scission and chain cross-linking occur simultaneously. However, at low electron and photon doses chain scission greatly dominates cross-linking and thus PMMA acts as a positive resist [28]. At significantly larger doses, free radical generation climbs steeply so that PMMA begins to cross-link profusely and thus begins to behave as a negative resist [29]. As we point out later, it is interesting that this tone reversal is seen with both electron and photon exposure.

Positive tone lithographic action can be obtained with pure PMMA films when irradiated with electron and ion beams, as well as with UV and X-ray photons. Indeed, these are the most well-understood lithographic processes with PMMA. Thompson and Kerwin wrote an early review on resist polymers for both photo- and electron beam lithography which is still very worthwhile to read [30]. Another similar exposition discusses the use of PMMA and other polymers as resists for ion beam exposure [31]. Recently, Gangnaik et al. have published a review that compares PMMA with newer generations of electron beam resists, such as ZEP [32]. They have shown comprehensively that while more modern resists may offer benefits of higher exposure sensitivity and enhanced resistance to dry etch plasmas, PMMA still wins hands down when it comes to sheer lithographic resolution. By extension, this also means that PMMA offers less line edge roughness (LER) when compared with other electron beam resists. While positive tonality is the one that is mainly used with electron beam, DUV and X-ray

exposures, negative tone action with pure PMMA films is only practical with electron beam irradiation. To obtain negative tone action with electromagnetic radiation requires the use of a suitable photo-cross-linking agent with PMMA [33], as is discussed later. This is because with photons, cross-linking does not dominate sufficiently over chain scission to provide a clean negative tone functionality with PMMA alone. As will be explained later, commercial photo-cross-linking agents are available that produce free radicals under suitable UV illumination to cross-link PMMA and thus deliver negative tone lithographic action. These types of resist compositions also use solvent-based developers which tend to dissolve away un-cross-linked PMMA, leaving patterned structures in the form of cross-linked PMMA regions. Structure in cross-linked PMMA can act as effective masks for dry etch-based pattern transfer processes. This is one area where PMMA has had a well-known weakness but as we discuss further on, this weakness (susceptibility to plasma erosion) can be overcome to a large extent through cross-linking of PMMA. Moreover, this protective cross-linking can be performed after either electron beam or optical lithography.

3. Positive tone lithography with PMMA

3.1. Positive tone lithography using electron beams

Polymethyl methacrylate (PMMA) has been used for several decades as one of the highest resolution electron beam resists known [34]. The linear structure of this polymer, together with favourable chain scission characteristics under electron beam irradiation, results in excellent post-development patterning characteristics [35]. PMMA has, therefore, been extensively used for fabricating ultra-small structures ranging from a few microns down to a few nano meters across in size [36–40]. It is interesting to note that the molecular weight of PMMA used for electron beam lithography has little bearing on the ultimate resolution that is achievable. This was experimentally verified by Khoury and Ferry by writing patterns on ultra-thin silicon nitride membranes [41]. Such membranes remove much of electron backscattering that is observed from bulk substrates and thus minimise proximity effect. This, then allows ultimate polymer chain length-dependent lithographic resolution limits to be ascertained. Later, Yan et al. also studied the dependence of lithographic resolution on PMMA molecular weight for pattern development at low temperatures [42].

In a typical electron beam lithography process, a thin (around 300 nm thick) PMMA film is spin coated on a flat substrate and then baked at around 90 °C for 15 min to remove any trapped solvent molecules. The coated substrate can then be patterned with a vector scan electron beam lithography system. Sometimes, these are dedicated purpose-built electron beam lithography tools but it is more usual to find modified SEMs being used for this purpose. A pattern is designed with a suitable computer aided design (CAD) package and the design file is used by the lithography system to deflect an electron beam, so as to trace out the pattern on the PMMA film. Energetic electrons (accelerated to energies somewhere in the range of 20 keV to 100 keV) scission long PMMA chains into smaller fragments and thus form a latent image in the film. Pattern contrast, thus, arises from the differing chain lengths of PMMA in exposed and un-exposed regions. Most electrons that expose PMMA are ones that are backscattered from the substrate after passing through the PMMA film. As backscattering disperses electrons into a wide cone so electron beam lithography suffers from 'proximity effect' where regions next to those that are intentionally exposed also get exposed to a certain extent. The detailed mechanism that gives rise to proximity effect in electron beam lithography has been described by Chang [43] whereas the minimization of proximity exposure through spatial dose equalization has been described by Parikh [44,45] and also by Owen and Rissman [46]. It is also possible to calculate the spatial distribution of backscattered electrons through Monte Carlo simulation techniques, as has been demonstrated by the IBM team of Kyser and Viswanathan [47].

Right after electron beam exposure, the exposed pattern may or may not be visible to the eye. A development step is generally carried out by dipping the substrate into a 1:3 mixture of methyl isobutyl ketone (MIBK) and isopropyl alcohol (IPA) [48–51], usually for half-a-minute to one minute duration. This step dissolves the shorter PMMA fragments and leaves almost all of the un-exposed PMMA film, thus creating a topographic pattern in the resist film.

Patterns lithographically formed in PMMA can be transferred to the substrate through either additive (metal deposition and lift-off) or subtractive (dry etch) processes. In practice, the latter is seldom used because of the poor dry etch resistance of PMMA. Lift-off processing for very small structures ($< 1 \mu\text{m}$) can pose problems due to material coverage on the resist side walls. This is made easier by the use of double-layer resist schemes where PMMA films of two different molecular weights are spun one on top of the other and then patterned through electron beam exposure. The top layer is higher molecular weight PMMA, compared to the bottom layer. With both layers getting the same spatial electron exposure, the bottom layer dissolves more quickly when developed on account of its lower molecular weight. This leads to the formation of an overhanging 'lip' or 'ledge' at the top of the resist openings which protect the resist sidewalls from getting covered by metal during subsequent metal deposition. This makes it easy for the solvent (usually acetone) to attack and remove the overlying PMMA film during metal lift-off. This process has been described in complete experimental detail for ultra-small structures by Rooks et al. [52] and for transistor 'T' gates by Ahmed et al. [53].

3.2. Positive tone lithography using X-rays

PMMA is also widely used as a positive tone X-ray resist. Even way back in 1980, Flanders reported the replication of 175-Å lines and spaces in PMMA using X-ray lithography [54]. Later, during the 1999–2002 period, Cerrina and Ade published detailed studies of PMMA and other organic materials as resists for X-ray lithography [55,56]. Another very useful survey, that also covers electron beam resists, has been provided by Schnabel and Sotobayashi [57]. At such short wavelengths, X-ray photons behave very much like accelerated electrons and thus X-ray lithography, except for the exposure source, works mechanistically very much like conventional electron beam lithography. However, the major difference is that whereas electron beam lithography is serial (line-by-line exposure), X-ray lithography is a parallel patterning process. This makes X-ray lithography of particular interest for ultra-small structure definition in the industry. The main challenge in carrying out X-ray lithography (other than having access to a suitable X-ray source) is having the right mask. X-ray lithography masks are significantly more challenging to make when compared with UV and DUV masks. Thus, a great deal of effort in X-ray lithography goes into making suitable masks which are generally made from thin suspended silicon nitride membranes overlaid with either gold or multilayer metal films of silicon and molybdenum; acting as X-ray blockers. Gold is generally preferred because due to its high atomic weight it provides good X-ray blocking. Molybdenum/silicon multilayers are used for reflective X-ray masks because they act as efficient Bragg mirrors at X-ray wavelengths. Good descriptive coverage in this area has been provided by Malek et al. They have described the requirements for deep x-ray lithography (DXRL) masks and also a cost-effective mask fabrication process. Furthermore, their review includes a summary of tabulated properties for materials used in the fabrication of DXRL masks [58]. Klein et al. have also described X-ray mask making techniques and their performance metrics with good detail [59]. X-ray sources, masks and PMMA as a resist for X-ray lithography are also particularly well described by Christenson and Guckel [60]. Their work was mainly carried out at the 2.5 GeV National Synchrotron Light Source (NSLS) storage ring at Brookhaven National Laboratory.

The PMMA used in X-ray lithography is, generally, developed by the same developer as is used for electron beam lithography i.e. a mixture

of MIBK and IPA. Some alternative developer schemes have been described by Malik and Yajamanyam [61]. Their studies of the development characteristics of PMMA under X-ray exposure included the influence of various parameters, such as resist thickness, pre-exposure thermal treatment, exposure conditions (x-ray incident energy, dose at the bottom of the resist, rate of dose deposition, distribution of dose inside the resist), development conditions (temperature, megasonic-assistance, length of cycle into developer and rinse baths), post-exposure treatments (storage, thermal treatment), etc. The study also focused on the surface finish of the parts corresponding to different exposure and development conditions.

Beyond conventional X-ray lithography, there have also been other innovations in this field. Sugiyama et al., for instance, have described X-ray exposure on a moving resist [62]. They employed X-rays from a Synchrotron source to expose a moving resist + substrate system. The resulting time-variable exposure was used to fabricate micro-lens and micro-needle arrays with sloped side-walls and curved surfaces, respectively. They named their technique plain-pattern to cross-section transfer (PCT) technique. X-ray lithography-based Micro-needle arrays have also been fabricated by Moon and Lee using inclined X-ray exposure [63].

3.3. Positive tone lithography using UV radiation

3.3.1. Use with DUV exposure

PMMA is only very weakly susceptible to chain scission with UV radiation at wavelengths longer than 300 nm. The propensity to chain scission at long UV wavelengths can, however, be increased by the incorporation of photochemical free radical generators, as will be discussed in the section on lithographic tone switching in PMMA. At UV wavelengths shorter than 300 nm PMMA undergoes sufficient chain scission that it can be used as a positive resist with strong (high intensity) UV sources. The mechanism of optically-induced chain scission is quite similar to that of electron beam-induced chain scission [64]. In this case, however, energy is transported to PMMA in the form of energetic photons. Absorption of DUV photons causes PMMA strands to vibrate violently and some eventually shear off into smaller fragments. The effectiveness of this process climbs steadily with reduction in UV wavelength. This is because of two reasons. First, shorter wavelength UV photons have a larger absorption coefficient in PMMA so their probability of interaction with PMMA chains is enhanced. This is readily evident from a spectral absorption plot of PMMA. Second, shorter wavelength UV photons carry enough energy to, every so often, scission a PMMA strand.

There are two different ways in which positive tone optical lithography can be performed with PMMA. One way is similar to conventional positive tone electron beam lithography and involves wet development after the exposure process. The developer is a weak solvent for PMMA which preferentially dissolves shorter PMMA fragments found in the UV-exposed regions. 1:3 MIBK:IPA is also a reasonable developer for this type of lithography. After exposure, the sample is developed in MIBK:IPA mixture to generate image topography in the polymer film.

A different, more interesting, way of carrying out positive tone lithography with PMMA involves heating the PMMA-covered substrate during DUV exposure. In this case, scissioned PMMA fragments leave the resist and thus no separate development step is needed after exposure. This technique, that we call dry lithography, requires the PMMA film to be heated to close to its glass transition temperature, T_g [65]. Dry lithography takes considerably longer to perform than conventional lithography with wet development because multiple chain scission steps are required to cut polymer fragments to very short lengths. Such short chain segments then volatilize at temperatures close to T_g . This process gradually thins down the PMMA film regions exposed to UV radiation while un-exposed PMMA retains very close to its original thickness. The patterning process is complete once all PMMA

under irradiated locations is removed to expose the underlying substrate. In this way, the PMMA film is sculpted to produce the same pattern that is present on the exposure mask [66]. As PMMA in UV-exposed regions is automatically removed through thermal volatilization so no separate wet development step is required. The advantages of dry lithography include vertical resist sidewalls and the ability to pattern even very thick PMMA layers; measuring up to several microns in thickness. This last advantage manifests because of the high UV transparency of PMMA which allows short wavelength UV photons to penetrate all through the thickness of the PMMA film. These attributes make dry lithography with PMMA attractive for such applications as the fabrication of microfluidic channels, micro-moulds, stamps for nanoimprint lithography (NIL) and micro-electro-mechanical systems (MEMs).

An interesting phenomenon that was observed in our work was the ejection of photoelectrons from the silicon substrate as DUV photons hit it, having traversed the overlying PMMA film [65]. At 254 nm, or shorter wavelengths, photons have enough energy to eject photoelectrons from silicon. These electrons cause a secondary exposure of PMMA from the substrate side. This has two effects: it increases the overall exposure of PMMA and it introduces a proximity effect to exposure. The latter, arises because the photoelectron ejection cone is somewhat wide and it opens up further at higher photon energies; enhancing the proximity exposure at shorter UV wavelengths. This effect becomes more pronounced as the wavelength of UV radiation is reduced. At the extreme ultraviolet (EUV) wavelength of 13.5 nm, slated for the coming generation of EUV lithography, photoelectrons and secondary electrons are the principal species that cause exposure of resist film [67]. Our work shows that even at 254 nm this process becomes very conspicuous. In fact, PMMA is so versatile as a resist material that it is also used at EUV exposure wavelengths where it offers superior resolution to chemically-amplified resists (which offer higher sensitivity to EUV radiation) [68,69]. The use of PMMA with readily available portable commercial EUV sources (see Fig. 2) enables EUV lithography to be performed at low cost, making this technique available to even academic laboratories.

Some tool modifications are needed to enable DUV lithography with

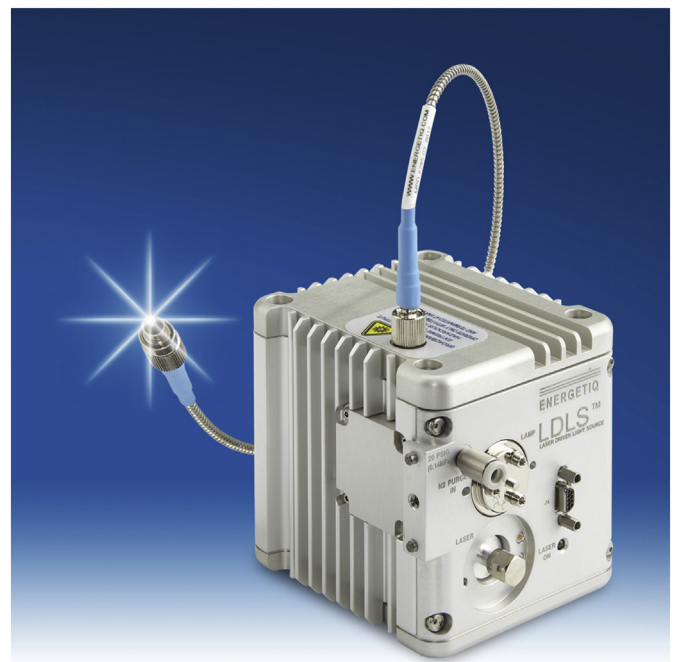


Fig. 2. Energetiq Technologies (Now, Hamamatsu Corporation) laser-driven light source portable EUV generator.

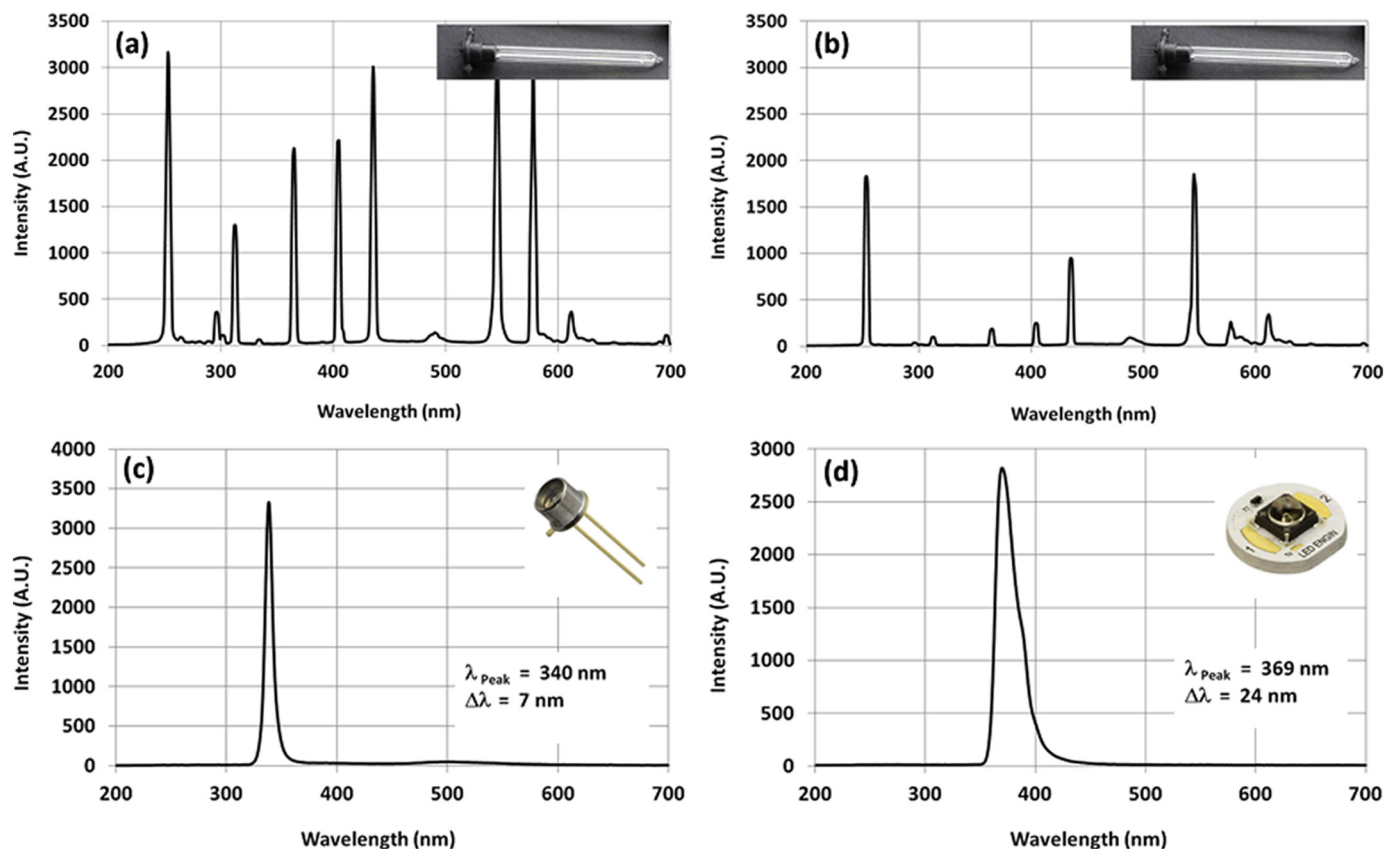


Fig. 3. Spectra of various discharge tube and LED sources useful for optical lithography with PMMA. Insets show the sources themselves. (a) high-power cold-cathode UV discharge tube, (b) low-power cold-cathode UV discharge tube, (c) 340 nm UV LED, (d) 369 nm UV LED.

PMMA. Ordinary mask aligners are designed with UV lamps and filters that make available 365 nm i-line radiation for lithography. DUV lithography, of the type described above, requires shorter wavelength radiation at 254 nm. High-pressure mercury vapour lamps used in mask aligners produce only small amounts of radiation at this shorter wavelength so different lamps have to be used, with all filters removed from the radiation path. Low-pressure mercury vapour lamps can be used for this purpose. We used in-expensive, low voltage DC-operated, quartz-envelope low-pressure mercury vapour lamps for our experiments (as seen in Fig. 3). Their only downside is that they emit relatively small amounts of radiation which necessitates long exposure times [70]. This figure also shows LED sources (with their spectra) that can be used for small-scale lithography experiments at longer UV wavelengths. A heated stage is also required if dry lithography is to be carried out. As for mask, ordinary chrome-on-glass and ferric oxide-on-glass masks are not suitable for DUV lithography because the glass absorbs a large fraction of DUV radiation. Masks using fused silica are needed and are easily available from most mask making shops at a somewhat higher price when compared with glass-based masks. If the pattern features are at millimetre scales then even a metal aperture shadow mask can be used. In fact, lithography with electro-etched metal shadow masks using thick PMMA resist films, followed by metal electro-deposition, can be used to form high aspect ratio metal structures with DUV dry lithography.

3.3.2. Use with i-line exposure

PMMA can also be used as a positive tone lithographic resist with exposure wavelengths above 300 nm. However, this necessitates the use of a chemical agent that extends the chain scissioning ability of PMMA to photons of longer wavelengths. A suitable solvent developer can then develop the pattern post-exposure. Photo-active free radical generators can serve in such a role. These reagents undergo light-induced bond

cleavage, generating a copious supply of highly active free radicals. If present in low or moderate concentration, these free radicals can attack and degrade PMMA, resulting in positive tone lithography. This has been well-described by Sugita, in his review of photo-degradation of resist polymers [71]. While a number of different photo-activated free radical generators suitable for these roles exist, we have extensively studied Irgacures, manufactured by the BASF Corporation, as enablers of photolithography with PMMA. Our studies have shown that two members of this family, Irgacure 651 and Irgacure 379, are particularly suitable for use as PMMA sensitizers. Their addition causes minimal disruption in the structural integrity of PMMA films while turning them into long wavelength UV-sensitive photoresists. Gałka's group has studied the photo-stability of PMMA containing photo-active molecules, including Irgacures, using optical spectroscopy. Their publications can be studied for details on how photo-cleavage of molecules such as Irgacures leads to fragmentation or cross-linking of PMMA [72,73]. Taking the case of Irgacure 379, its addition to PMMA turns the acrylic polymer into a strong UV absorber. Low concentrations of this photo-activated free radical generator ($\leq 5\%$, w/w) greatly enhance UV-induced chain scission (fragmentation) of PMMA. This results in positive tone lithographic action. However, the contrast is quite low which means that long development times are required and the resist film gets thinned down considerably by the time resist in exposed areas is completely removed. As a result, the patterned resist film left behind after UV exposure is too thin to be of much value for either additive or subtractive pattern transfer. For this reason, this mode of action is not of practical importance. Fig. 4 shows a pattern created in 996,000 g/mol molecular weight PMMA containing $\sim 5\%$ (w/w) Irgacure 379 after i-line exposure in a mask aligner and wet development with 1:1 MIB-K:IPA developer. The dark portion is the intact (but thinned down to around 72 nm thickness) PMMA resist film whereas the light-coloured line patterns are locations where the resist film has been completely

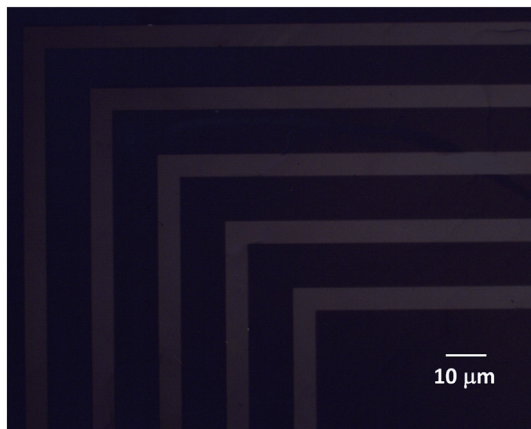


Fig. 4. Positive tone lithography pattern in PMMA containing 5% Irgacure 379 after exposure with i-line radiation and wet development with MIBK:IPA developer.

removed due to the fragmentation of PMMA, such that the underlying silicon substrate has become visible.

4. Negative tone lithography with PMMA

4.1. Negative tone lithography using electron beams

While PMMA excels for defining nano-scale features, it has poor erosion characteristics under typical dry-etch plasmas used for subtractive pattern definition. In fact, typical i-line photoresists have comparatively better dry-etch plasma resistance because of the presence of phenolic groups in their polymer matrix [74]. The etch resistance of PMMA can be increased considerably by cross-linking it with high doses of electron beam radiation i.e. using PMMA as a negative tone electron beam resist [75]. Using negative instead of positive tone mode of lithography requires a very large increase in electron beam exposure dose. The significantly larger dose causes excessive production of free radicals which lead to overall cross-linking of the PMMA film. Cross-linking causes the plasma etch resistance of PMMA to increase and thus this technique has been used in the past to etch more deeply into underlying substrates than would be possible with non-cross-linked PMMA [76–79]. The downside of this approach is a loss of resolution because cross-linked PMMA tends to swell during development with organic solvents, causing patterns to get distorted. This happens because solvent molecules get trapped in the dense multiply-bridged matrix of cross-linked PMMA, causing gel formation and consequent increase in bulk volume. Papanu et al. studied the amount of PMMA swelling during wet development using ellipsometry while using organic solvent developers of different molecular weights and different development temperatures [80]. Low molecular weight solvents diffuse somewhat more easily into PMMA and thus cause more swelling and pattern diffusion. Similar findings have also been described by other research groups [81,82]. Some resolution is also lost due to the parasitic effect of higher backscattering on account of higher irradiation dose needed for polymer cross-linking. Similar behaviour is seen in other negative tone organic resists, such as polyethylene glycol (PEG) [83].

Performing electron beam lithography on highly resistive substrates could pose problems due to lack of sufficient charge dissipation. In such cases, where sample charging and consequent beam deflection-induced pattern distortion can be a problem, a thin (~100 nm thick) layer of electron beam-transparent polythiophene can be spin coated on top of the PMMA film to dissipate the charge. As aqueous dispersions of polythiophene are commercially available so spin coating a water-based polythiophene solution on PMMA poses no problem because PMMA itself is insoluble in water. This method has been shown to be very effective for lithography on such resistive materials as zinc oxide,

gallium nitride and diamond [84,85].

4.2. Negative tone lithography using UV radiation

It has been stated above that low concentrations of Irgacure 379 cause PMMA to fragment on exposure with 365 nm UV radiation. With Irgacure 379 or Irgacure 651 concentration in excess of 5%, however, PMMA undergoes substantial cross-linking when exposed to i-line radiation. PMMA + Irgacure system then behaves as a negative tone resist [86]. The amount of cross-linking thus produced climbs with increase in the amount of Irgacure 651 added to PMMA. An optimum concentration of Irgacure 651 appears to be between 15% and 20%. Maximum useful UV sensitivity is observed in this range. Beyond 20% Irgacure 651 in PMMA, films begin to lose structural integrity and tend to peel off from the substrate, as well as crack during the development process. Irgacure 651-containing PMMA acts as a fairly good negative tone optical resist which can be used at the i-line wavelength (365 nm) [87,88].

As elaborated further in Section 6 below, increase in Irgacure concentration leads to excessive production of free radicals. These free radicals then react with PMMA chains to form inter- and intra-chain cross-links. Free radical concentration can also be boosted by simply increasing the UV radiation dose provided to pure PMMA. But to increase their concentration to the extent that it results in PMMA cross-linking, simply by increasing the UV radiation exposure, would require an impractically large UV dose. Irgacure thus performs two useful functions: it greatly lowers the dose requirement for cross-linking of PMMA and also makes PMMA sensitive to 365 nm UV radiation. The latter can be seen in Fig. 5 where the transmission spectra of pure PMMA and PMMA containing 15% Irgacure 651 have been plotted together. Strong UV absorption in the sub-400 nm region is clearly evident. Patterns can be lithographically formed in the PMMA + Irgacure resist system through i-line exposure using ordinary mask aligners. Right after exposure, changes in PMMA chain lengths create sufficient refractive index contrast that the exposed regions can be distinguished from un-exposed regions; as seen in the inset to the right in Fig. 5 (unexposed: greenish region at the top, exposed: reddish region at the bottom). A wet development step using MIBK diluted with IPA is required after exposure to form actual relief structures in the resist film. The right inset to Fig. 5 shows an AFM micrograph of part of a 10 μm line – 10 μm space grating structure formed in PMMA through this

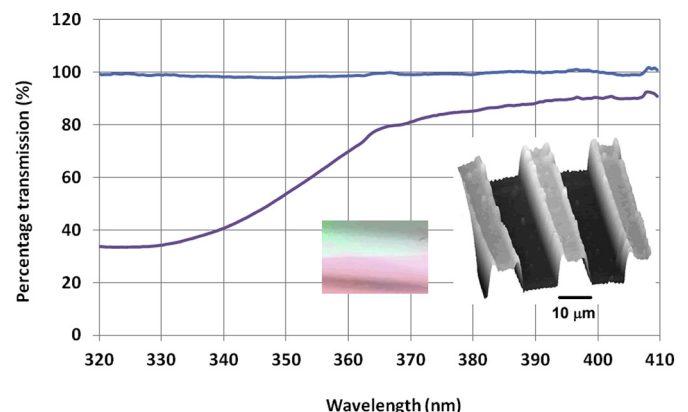


Fig. 5. Percentage transmission of UV radiation through a 1 μm thick PMMA film on quartz substrate (top blue line). When 15% (w/w) Irgacure 651 was added to PMMA then there was clear UV absorption (lower purple line). Inset at the left shows a resist sample that was half exposed to UV radiation at the bottom. Showing visible colour contrast after exposure but prior to wet development. Inset at the right shows the topographic structure of a section of a grating formed by negative tone optical exposure, as revealed by atomic force microscopy. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

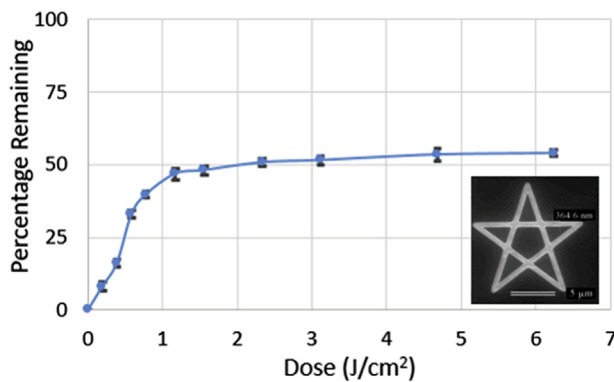


Fig. 6. Thickness vs. 365 nm exposure dose for PMMA containing 20% Michler's ketone as a photo-active cross-linking agent. The inset shows a star pattern formed in this resist with positive tone electron beam lithography.

process. Afterwards, the resist pattern can be transferred to the underlying substrate using a subtractive pattern transfer processes.

Another photo-activated cross-linking agent that can be used for this purpose with PMMA is Michler's ketone. This organic compound is a derivative of benzophenone with the formula: $[(CH_3)_2NC_6H_4]_2CO$ and has the formal name bis[4-(dimethylamino)phenyl]methanone. It absorbs radiation strongly at 366 nm – almost at the i-line wavelength. This compound is commercially available from several sources as a white powdered solid. We have found that PMMA containing 20% Michler's Ketone can act as both a cross-linkable resist under i-line exposure and as a positive and negative resist with electron beam exposure. Fig. 6 displays a plot that shows increasing cross-linking in this resist after exposure to increasing doses of 365 nm UV radiation. As can be seen, for optical exposure doses above 1 J/cm^2 the resist shows almost full cross-linking. The inset depicts a star pattern formed in the same resist (PMMA containing Michler's Ketone) with positive tone electron beam lithography. This shows that this resist can work as both an electron beam resist and as an optical resist.

An interesting observation with the use of Michler's ketone for negative tone lithography was the formation of a 'lip' structure along open features (see Fig. 7). This very likely arises because of strong UV absorption by Michler's ketone so that the top parts of exposed PMMA get significantly more cross-linked compared with the bottom of the film. During development, the lower portion of the film dissolves somewhat more, creating the lip-like structure seen in Fig. 7.

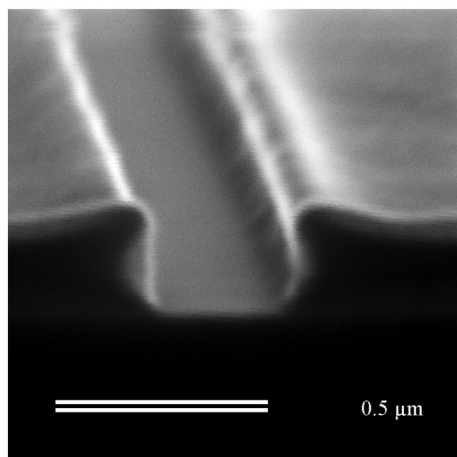


Fig. 7. Lip formation along open features in negative tone pattern formed in PMMA, using Michler's ketone.

5. Topographic structure formation in PMMA using resist reflow

Exposure of PMMA to electron beam at dose levels suitable for positive tone lithography creates shorter chain PMMA strands that generally extend throughout the PMMA film down to the substrate. This exposed region has physical properties that are somewhat different from that of the surrounding un-exposed region. Shorter chain PMMA has somewhat lower refractive index and this sometimes makes exposed regions visibly stand out in contrast with the surrounding un-exposed region. Shorter polymer chains also are more mobile when compared with longer chains of the same polymer. This has important implications for a PMMA film where chain length contrast has been created through patterning with electron beam exposure.

An electron beam-patterned PMMA film, if heated to close to the glass transition temperature, held at that temperature for some time and then cooled back to room temperature, develops a topographic (height) contrast. The amount of time needed for this depends on the sample temperature with less time required at higher temperatures. Temperatures too close to the glass transition temperature, however, can cause the pattern to distort and are, therefore, not advisable. Such post-exposure thermal treatment usually makes the pattern clearly stand out because now there is both a refractive index contrast and a film thickness contrast. While the former derives from the lower refractive index of shorter PMMA chains, the latter is due to the loss of material from exposed regions after thermal treatment. There are two prominent causes for material loss that is observed after post-exposure thermal bake. Very short PMMA chains and molecular fragments get lost through a sublimation process; both during electron beam exposure and during the subsequent thermal treatment. Somewhat longer chains, which have lower T_g and are significantly more mobile, simply dissolve into the surrounding matrix of un-exposed, and thus long chain, PMMA [89]. Together, these effects lead to a considerable loss of material from the exposed regions and, consequently, the film thickness is significantly reduced in these regions. The overall effect is that of topographic variation throughout the PMMA film, with sharp features becoming rounded due to PMMA reflow. Obviously, this technique can be used to fabricate three dimensional structures in PMMA. Fig. 8 shows an array of micro lenses fabricated using this technique. An array of convex micro-lenses is seen on the left (a) whereas an array of concave micro-lenses is seen to the right (b). In contrast to the e-beam exposure process, both of these micro-lenses were formed through a 3 h DUV irradiation process with an optical power of 5.8 mW/cm^2 . This exposure process is very similar to the e-beam process as both cause chain scissioning to occur upon exposure. After the exposure, the samples were given a 30 min 150°C post-exposure bake. Finally, various parameters of the micro-lenses were determined. The convex micro-lens diameter, height and focal length were determined to be $140 \mu\text{m}$, $9.23 \mu\text{m}$ and $563 \mu\text{m}$, respectively, while the concave micro-lens diameter, height and focal length were determined to be $54 \mu\text{m}$, $2.9 \mu\text{m}$ and $-261 \mu\text{m}$, respectively.

It should be noted here that refractive micro-lenses are harder to fabricate than Fresnel micro-lenses because the former require smooth three-dimensional structures. Indeed, Fresnel micro-lenses fabricated using electron beam lithography were reported far back in the 1980s [90] but refractive micro-lens fabrication has only been reported relatively recently.

6. Lithographic tone switching in PMMA brought about by UV exposure

As mentioned in the sections above, PMMA changes from a positive tone electron beam resist to a negative tone resist at high electron beam dose. This transition takes place over an extended dose range which depends on such factors as the molecular weight of PMMA, its thickness, substrate composition and its density. The change in lithographic tone is driven by a gradual change in PMMA's tendency from chain

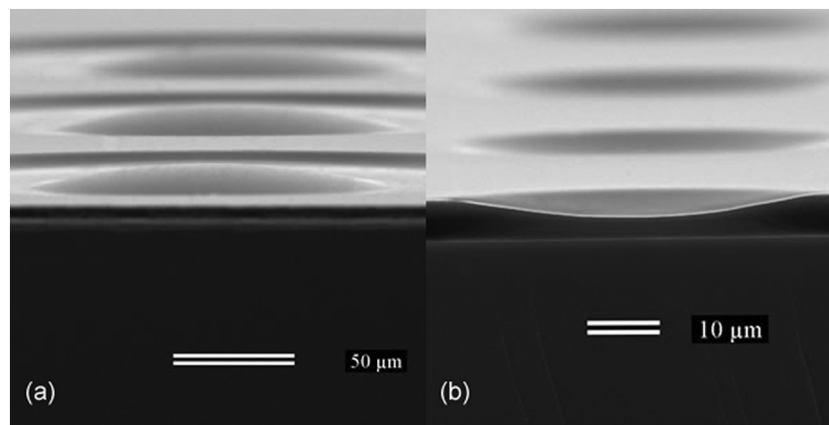


Fig. 8. Array of convex lenses (a) and concave lenses (b) formed by a PMMA resist reflow process.

scission to cross-linking as the electron beam dose increases [91]. Ultimately, this effect arises because of an increase in free radical concentration as an incident electron beam interacts with PMMA molecules. Shearing off of polymer backbone and side chains due to electron impact generates both molecular fragments and free radicals. At low concentration, free radicals aid chain scission but at high concentrations they begin to generate inter- and intra-chain cross-links and then the balance of reaction shifts toward cross-linking. Thus, some amount of chain scission is always present though at significantly high free radical concentration more cross-linking than chain scission takes place so that the polymer begins to gain molecular weight rather than losing it. This dose-dependent tone switching behaviour can be utilised for specialized fabrication requirements [92].

While PMMA's lithographic tone switching behaviour with change in electron beam dose is well known, our work has shown that PMMA behaves in much the same way with electromagnetic radiation as well. However, with UV radiation the change in lithographic tone takes place over a much wider dose range. Added to that the fact that high enough UV dose, sufficient to cause cross-linking in PMMA, will impart so much energy to the polymer that it will likely ablate, makes it impractical to cross-link PMMA directly with UV exposure. Our approach to demonstrate UV radiation-mediated lithographic tone switching in PMMA was through adding Irgacure 651 to it. As mentioned before, Irgacure amplifies free radical concentration in PMMA by generating an additional supply of free radicals beyond that generated through PMMA chain fragmentation. Our experiments have shown that at low Irgacure 651 concentrations of around 5% (w/w) the additional free radicals, generated through photo-cleavage of Irgacure 651 molecules, make PMMA greatly amenable to chain scission. Thus, PMMA containing 5% Irgacure 379 or Irgacure 651 is an i-line-sensitive positive tone optical resist. However, when we increased the concentration of Irgacure 651 to any value in the range of 5% to 15% then PMMA, at large 365 nm exposure dose, was found to cross-link instead. Increasing the amount of Irgacure 651 incorporated into PMMA also increased the degree of cross-linking for the same 365 nm UV exposure dose but going beyond 15% Irgacure 651 concentration started to negatively affect the structural integrity of the resist film. As increasing the concentration of Irgacure 651 increases free radical concentration in PMMA in much the same way as an increase in UV exposure dose so this indirectly demonstrates lithographic tone switching from positive to negative in PMMA with 365 nm UV exposure, using Irgacure 651 as an optically-activated free radical generator.

In a more direct fashion, we also found that at 15% Irgacure 651 concentration PMMA exhibited chain scission at low 365 nm UV radiation dose (below 1 J/cm^2) but cross-linking at much higher UV exposure dose. The transition from positive to negative behaviour in this manner is clearly seen in Fig. 9 where we have plotted the percentage thickness of PMMA + Irgacure 651 resist film remaining after wet

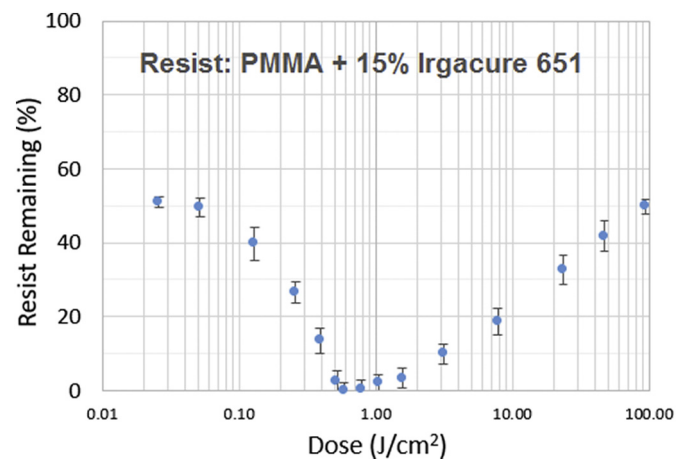


Fig. 9. Transition from positive to negative tone lithography through optical exposure of Irgacure 651-containing PMMA.

development in MIBK at room temperature against the UV exposure dose. Each data point shown was obtained with a separate resist film and its corresponding exposure and development. Post-development, PMMA film thickness first decreases indicating positive tone action but at much larger doses it begins to recover, indicating negative tone action. The use of logarithmic scale for UV exposure dose highlights the fact that very large changes in UV exposure dose are required to see the transition from the positive tone (chain scission-dominated) to negative tone (cross-linking-dominated) regime. To understand Fig. 9, please note that the film height did start from 100% initial resist thickness. It is simply because here the data is presented on a logarithmic scale and does not start from 0 (J/cm^2) exposure dose that it seems that the film height starts from 50%. This optical tone switching behaviour, so succinctly illustrated in Fig. 9, exactly parallels the well-known lithographic tone switching seen in PMMA under electron beam exposure.

7. PMMA as a combined optical and electron beam resist

From discussions in the sections above, it is clear that PMMA can be used not only as an electron beam resist in both positive and negative tone mode but also as a UV resist at both long (negative tone) and short (positive tone) wavelengths. Our investigations have shown that the processes of electron beam and optical lithographies can even be combined on the same PMMA film [93]. Please note here that we used 20 kV electron accelerating voltage in all of our work described in this paper. The great utility of this combination lies in the fact that it allows structures to be patterned at both small and large length scales using electron beam and UV lithography, respectively, on the same resist film.

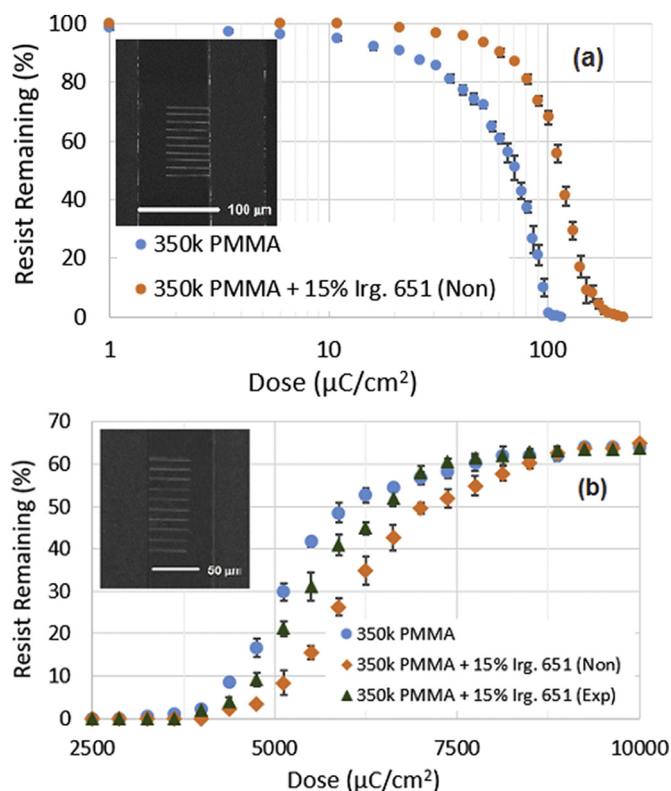


Fig. 10. (a) Dose plot for positive tone electron beam lithography with 350,000 g/mol PMMA. Inner set of blue points show the percentage height of PMMA remaining after wet development with MIBK:IPA developer at different exposure doses. Outer set of orange points are similar data for PMMA containing 15% of Irgacure 651. (b) Same as above at much higher exposure doses that cause negative tone action. Here, the green triangles are data points for 15% Irgacure-containing PMMA, after UV exposure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

This is made possible by the fact that adding Irgacure to PMMA does not impair the latter's capability to act as an electron beam resist. This can be seen in Fig. 10 which shows dose plots for positive and negative tone electron beam lithography with 350,000 g/mol PMMA. In part (a), the inner set of blue points show the percentage height of PMMA remaining after wet development with MIBK:IPA developer at different exposure doses. The outer set of orange points presents similar data for PMMA containing 15% of Irgacure 651. Notice that the presence of Irgacure 651 slows down the resist action i.e. requires a larger exposure dose. Thus, where 100 $\mu\text{C}/\text{cm}^2$ dose clears off pure PMMA (0% resist film height remaining after wet development), to achieve the same with PMMA containing 15% (w/w) Irgacure 651 requires a dose close to 200 $\mu\text{C}/\text{cm}^2$. We believe this increase in dose requirement arises due to extra electron scattering in PMMA containing Irgacure 651 molecules.

Part (b) of Fig. 10 is a similar dose plot but for much higher electron beam doses that result in negative tone action. Again, the set of blue points show the percentage height of PMMA film remaining after wet development with MIBK:IPA developer at different exposure doses. Orange diamonds show similar data for PMMA containing 15% (w/w) Irgacure 651. It can be seen that the latter requires a higher dose when compared with the former. Where pure PMMA requires about 7500 $\mu\text{C}/\text{cm}^2$ to leave 60% of film thickness intact after wet development, Irgacure 651-containing PMMA requires close to 8500 $\mu\text{C}/\text{cm}^2$ for the same film thickness. We also carried out the same experiment after subjecting an Irgacure 651-containing PMMA resist film to a prolonged UV exposure at 365 nm. This 'optically-exposed' film behaved in a similar way as un-exposed film but required less dose for comparable

exposure with electron beam than an un-exposed film, as can be seen in part (b) of Fig. 10 (green triangles). We attribute this to photo-cleavage of Irgacure 651 which produces smaller molecular fragments that cause less scattering of incident electrons. These results clearly show that PMMA with added Irgacure 651 continues to work as positive and negative tone electron beam resist although with higher dose requirement. The presence of Irgacure 651, however, also enables the resist to be used as an optical resist.

In a combination lithography process two sequential exposures are carried out on the same PMMA film. Usually, optical exposure is performed first, followed by electron beam exposure. The UV exposure, carried out through either a UV-transparent optical mask or a metal shadow mask, can be used to define larger patterns whereas the following electron beam exposure can be used for delineating smaller patterns. After both exposures, a single resist development step is carried out to selectively remove PMMA, leaving a patterned film with features at both large and small length scales. The structures can then be transferred to the underlying substrates through either additive or subtractive technique in a single process step.

Because both optical and electron beam lithographies with PMMA can be performed in either positive or negative tone mode so altogether there are four distinct ways in which combination lithography processes can be carried out. These can be indicated as $+/+$, $+/-$, $-/+$ and $-/-$ where the first sign indicates the tonality of optical (UV) exposure and the second sign indicates the tonality of electron beam exposure. All four combinations are of utility for patterning different types of structures.

The $+/+$ process involves short wavelength UV exposure of PMMA in areas that are to be removed, followed by a positive tone electron beam exposure in areas not previously exposed by UV radiation. Each exposure results in chain scissioning of PMMA with the scissioned areas removed during the unified wet solvent development stage. The doses for optical and electron beam exposures are chosen such that an equal amount of chain scissioning takes place in both optical and electron beam exposed regions. In this way, both patterns get developed together during the wet development stage after exposures have been completed. The inset to Fig. 10 (a) shows a $+/+$ pattern where a large vertical grating was first defined by optical exposure and then a 250 nm line width grating was written on the same resist film using electron beam lithography. Both patterns developed together at the wet development stage.

The $+/-$ process is carried out with a short wavelength UV exposure followed by a high dose electron beam exposure in areas that have already been exposed by DUV radiation. In this way, the DUV exposure first produces chain-scissioned regions within which some smaller areas are cross-linked through electron beam exposure. During wet development, chain-scissioned PMMA is removed but cross-linked PMMA nested inside chain-scissioned regions is left behind.

The $-/+$ process involves the combination of negative tone 365 nm UV lithography with positive tone electron beam lithography. High dose UV exposure is first used to create optically-cross-linked PMMA region and then a subsequent electron beam lithography step is used to open up finer features in cross-linked PMMA. This is possible because electron bombardment on even cross-linked PMMA can fragment the chains and lead to the reappearance of solubility in developer solvents. An example of this appears in the inset to Fig. 10 (b).

Combination lithography can be performed using both Irgacure family of photo-active free radical generators and other suitable light-activated photo-cleavable moieties, such as Michler's ketone. Fig. 11 summarizes the various lithography possibilities with PMMA as a resist polymer. It should be mentioned here that the idea of combining optical and electron beam lithography on the same resist film is not new [94,95] but putting PMMA to this use is extremely convenient on account of its widespread usage, extremely high resolution and other desirable properties.

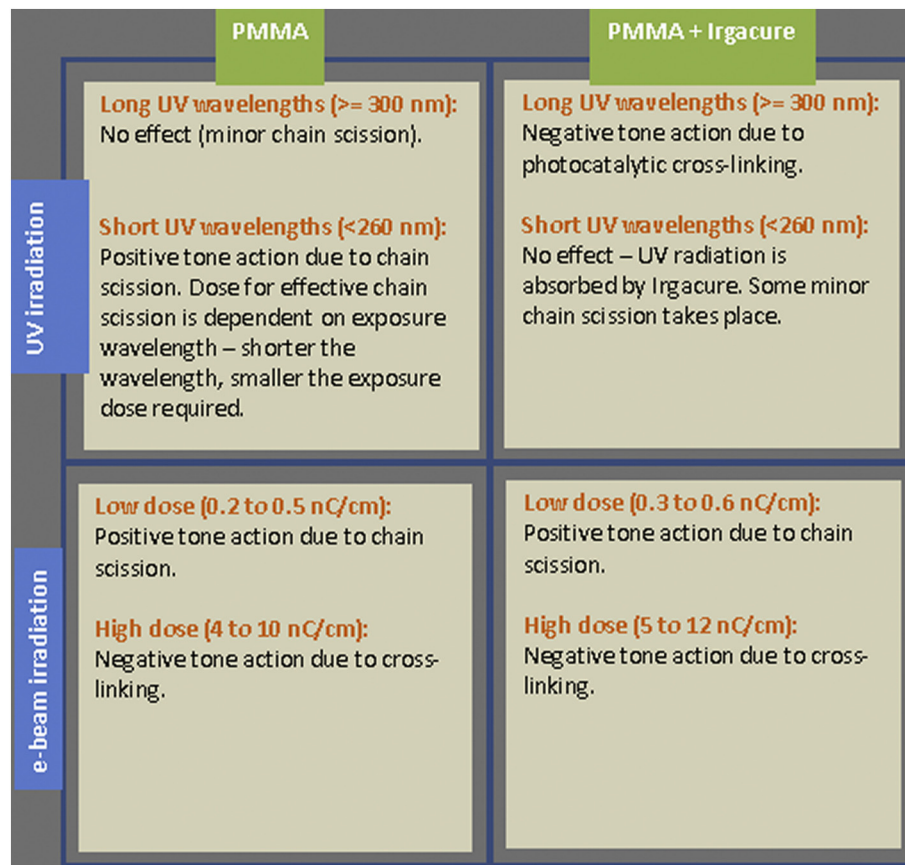


Fig. 11. Diagram summarizing the behaviour of PMMA in response to exposure with electron beams and UV radiation at different dose levels.

8. Dry etch resistance of PMMA

PMMA is well known to have very limited resistance to attack from dry etch plasmas [96,97]. This fact is particularly troublesome for this otherwise excellent resist, as far as patterning resolution is concerned. Using gel permeation chromatography on PMMA samples that underwent etching in CF_4/O_2 plasma, Harada has shown that PMMA undergoes random chain scission at both the surface and the interior, during plasma etching. Neutral and ionic etchant species permeate inside PMMA films to cause such structural damage [98]. The inferior dry etch resistance of PMMA ultimately arises from its paucity of carbon atoms and its relative richness with hydrogen and oxygen atoms. These, together with the open chain aliphatic structure of PMMA, make it especially susceptible to degradation under bombardment with energetic particles and photons – typical constituents of all dry etch plasmas [99–101]. The presence of aromatic rings, such as phenyl groups, in a polymer, on the other hand, enhances resistance against plasma erosion [102]. A resist should ‘resist’ chemical and physical degradation during etch processes, as much as possible but, unfortunately, PMMA falls far short in this regard. Our work has demonstrated that the ability to cross-link PMMA, as described above for negative tone optical lithography, can also be utilised, with advantage, to enhance the dry etch resistance of PMMA, after it has been patterned with electron beam lithography.

We have convincingly demonstrated, as described above, that when Irgacure photo cross-linking agent is added to PMMA then the photo cross-linkable resist thus obtained retains most of the resolution capability of pure PMMA under electron beam exposure [103]. Thus, Irgacure-added PMMA can be patterned, as usual, with an electron beam lithography system (using the usual 1:3 MIBK:IPA developer). After the patterning step, the patterned resist film can be cross-linked through exposure with 365 nm UV radiation. This hardens the pattern-carrying

polymer film such that it becomes significantly more resistant to erosion from dry etch plasma when compared with a pure PMMA patterned film. This process is very easy to implement and can make a great deal of difference when significant etch depths are required with deep sub-micron patterns. In comparison, an ion beam-based post-exposure cross-linking process described by Mochiji et al. is cumbersome to carry out because of its need for specialized ion beam exposure equipment [104].

The conclusion from our work was that 15% Irgacure 651 (w/w) was the best additive for 960,000 g/mol PMMA, for this application. Irgacure-added PMMA then retained most of the resolution capability of pure PMMA but became capable of strongly cross-linking into a much harder mask on UV exposure after electron beam patterning. This resist system has been shown to work optimally with a line dose of 0.48 nC/cm which is 77.8% higher than the 0.27 nC/cm line exposure dose for pure 996,000 g/mol PMMA. Thus, Irgacure 651-containing resist requires a substantial increase in exposure dose over pure PMMA. At the same time, diffused scattering of electrons in Irgacure 651-containing resist also cause the line widths to increase by about 8%. Thus, the dry etch resistance benefits conferred by Irgacure 651 come at the expense of increased electron beam dose and increased line widths. These increases, however, are not prohibitive and easily allow sub-100 nm features to be replicated in this resist system.

Fig. 12 shows an atomic force microscope image of a star pattern etched about 120 nm deep into silicon using post-lithography UV-hardened PMMA resist film. Much deeper etches are possible when using somewhat thicker resist films. This is an easy technique to implement when fine feature patterning is needed, together with the ability to etch deeper into the substrate than would be possible with pure PMMA alone.

It should be mentioned here that in all instances where PMMA is cross-linked, whether through UV exposure or through electron beam

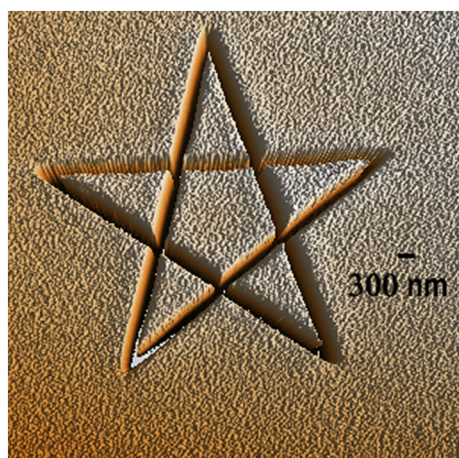


Fig. 12. Atomic force microscopy (AFM) image of star pattern dry-etched into silicon from cross-linked PMMA resist which was patterned by electron beam lithography and then exposure-hardened using 365 nm UV radiation.

exposure, the resist mask becomes insoluble in organic solvents. Any remaining mask after pattern transfer has to be removed through an oxygen plasma ashing process.

9. Conclusions

We have described a broad overview of the utility of PMMA as a resist for micro and nano fabrication. The discussion shows that this versatile polymer can be used for lithography in ways that are more extensive than is generally appreciated. PMMA is a well-known high-performance resist used for mainly positive tone electron beam lithography. By changing the exposure dose it can also be used in negative tone mode. Its derivatives also form the basis of resists used for DUV lithography. What is less well-known is that PMMA can also be used, as a resist, with optical exposure in the longer wavelength UV region. By changing the exposure dose, cross-linking additives, process temperature, development conditions and post-exposure processing, it is possible to get a variety of patterning possibilities. DUV exposure alone can lead to positive tone optical lithography whereas longer wavelength i-line exposure in the presence of cross-linking agents can result in negative tone lithography. Structures formed through positive tone optical lithography in PMMA can be 'reflowed' to create three-dimensional relief structures with rounded corners. It is even possible to combine both electron beam and optical lithography on the same PMMA film to obtain a range of patterning effects. Finally, the incorporation of a cross-linking agent, such as Irgacure 651, can be used as a way to cross-link PMMA after electron beam patterning and thus confer greater etch resistance to the patterned resist during plasma etching. Through a deeper understanding of the behaviour of PMMA it is possible to broaden the scope of lithographic patterning processes. This would ultimately lead to a better ability to handle challenging microfabrication tasks.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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